ASVIN ANG CHIT SIK

PMO Director · Project Management · Strategy · Digital Infrastructure · AI · Digital Transformation

English (Native) | Malay (Proficient) | Mandarin (Intermediate, spoken)

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EXECUTIVE SUMMARY

A senior leader with 18 years of experience across civil engineering, infrastructure development, and digital transformation — with a consistent record of building the tools and systems that others rely on to make decisions and deliver outcomes.

Uniquely positioned at the intersection of **deep domain expertise** and **applied AI**: having designed transport masterplans, delivered data centre pre-development programmes across APAC, and independently built geospatial platforms, multi-agent AI pipelines, and PMO automation systems using Python, React, and large language models.

Equally effective as an engineer, a strategist, and a builder — **the person who understands the problem, designs the system, and ships the tool**.

CORE COMPETENCIES

AI & Intelligent Systems

Multi-agent pipeline design
LLM integration (Claude, OpenAI)
Geospatial AI & automated analysis

Digital Transformation

Enterprise digitalisation strategy
Legacy-to-cloud migration
IoT, sensors & smart infrastructure

Programme & PMO Leadership

End-to-end programme architecture
Executive reporting & governance
Cross-functional team leadership

Infrastructure Development

Data centre pre-development (APAC)
Industrial park masterplanning
Site selection & technical due diligence

Geospatial & Data Platforms

Full-stack geospatial web apps
QGIS automation & spatial analysis
Open data pipelines (OSM, government)

Engineering Leadership

Civil, transport & hydraulic design
Procurement & contract management
Multi-disciplinary technical oversight

ABOUT

I turn ambiguous, undefined problems into frameworks and tools that organisations actually use. My career has moved across civil engineering, real estate development, industrial park operations, and data centre pre-development — not by accident, but because the same core skill applies everywhere: take a messy system, understand it fully, then build something better.

I approach every role with a systems-first mindset. I don't just fix a problem once — I build the platform so the problem doesn't come back. This has meant designing PMO frameworks, geospatial web applications, multi-agent AI pipelines, and enterprise automation ecosystems, often in parallel with running the day-to-day operations they were built to replace.

PROFESSIONAL EXPERTISE (YEARS)

WORK EXPERIENCE

Mar 2025 to current

PMO Director

Zerra DC, SG

Roles and Responsibilities

- **Leads the Corporate PMO function** (Legal, Customer Success, PMO) for a data centre developer in Singapore, reporting to the CEO and executive leadership team.
- Acts as **Strategic Integrator** across commercial, design, development and finance — bridging high-level strategy with regional execution in a startup environment.
- Designs and governs the delivery operating model, including governance frameworks, reporting cadence, digital platforms and cross-functional tooling.

Achievements

- **Established the E2E Asset Delivery Model** with 7-stage formal gates and standardised RACI — institutionalising accountability across the development lifecycle.
- **Designed and implemented the Smartsheet ecosystem** and migrated functions from static Excel to a live API-integrated database; launched Confluence as the central knowledge and AI tooling hub.
- Co-developed the **Elemental Cost Model** for standardised capex planning, bridging finance and project costing functions.
- Coordinated the **full rebranding transition from AGPDC to Zerra DC**, developing presentation decks, collateral, templates and internal materials.
- Built the **tender documentation for Project Dune**, supporting the Development Team in preparation of the GC tender package.

Apr 2023 to Mar 2025

Senior Program Manager

AirTrunk (Blackstone), SG

Roles and Responsibilities

- **Led and owned the End-to-end Pre-development Programme** for 30+ data centre projects across Singapore and Malaysia, reporting directly to the CEO and executive leadership team.
- Designed and governed the pre-flight framework: orchestrating site selection, design, construction readiness, customer engagement, and utilities across internal functions.
- Independently built **AI-powered tools and automation systems** that improved programme visibility, accelerated research, and reduced manual reporting overhead.

Achievements

- **Designed and rolled out the End-to-end Programme Management Framework** adopted across 30+ pre-development projects — covering gates, templates, reporting cadence and risk governance.
- Delivered pre-construction programmes for **4 data centre projects in Singapore and Malaysia** on schedule.
- Built a **nine-agent AI market entry system** (Claude API) producing fully branded research reports covering power grid, land, regulatory, connectivity, climate risk and competitive landscape — cutting research time from weeks to hours.
- Developed **Emerra.io** — a browser-based geospatial platform for infrastructure site suitability analysis using open data, constraint scoring and automated red flag reporting (React, FastAPI, GeoPandas, Firebase).
- Built a **Smartsheet-to-Confluence PMO automation suite** — syncing project data, auto-generating narrative status reports, and feeding executive dashboards via Power Automate and Python.

Nov 2021 to Apr 2023

Technical Manager — Land Acquisition (APAC)

Yondr Group, SG

Roles and Responsibilities

- Reported to the Head of Global Development Tech team.
- **Led the land acquisition technical due diligence activities** (fatal flaw analyses, site due diligence, engagement of consultants) across APAC, and **supported the Development team in coordination of investment negotiations** (site investigation, condition precedents for purchase and lease agreements).

Achievements

- Provided technical support in the assessment and **acquisition of a 300MW DC site in Johor, Malaysia**.
- Completed due diligence studies across Japan, Malaysia, Indonesia and Thailand on schedule and within budget.
- Developed consultancy procurement process including template scope documents, agreements and deliverables.
- **Redesigned and implemented a risk-based Fatal Flaw Analysis** framework based on ISO 13000.
- **Developed an automated QGIS site assessment tool** cutting DC site evaluation from 3 hours to 15 minutes.

Dec 2020 to Nov 2021

Senior Manager — Deputy Head, Design & Construction

Sembcorp / Kendal, ID

Roles and Responsibilities

- **Deputy Head of Design and Construction**, overseeing Planning, Procurement and Construction departments.
- Head of Planning (team of 6) and Head of IT (team of 5), reporting to the President Director.

Achievements

- **Spearheaded the park's digital transformation**: IoT sensors, smart monitoring, mobile tenant app, QR-based asset management system, and full cloud migration from physical servers.
- Developed 1,000ha expansion masterplan including **strategy for acquisition of 300+ parcels** with costing and development schedule.
- **Delivered USD 500k savings** through commercial negotiations for a 24,000m³ Water Treatment Plant and 16,000m³ Sewage Treatment Plant.
- Assisted in negotiations of a **120MW power supply agreement** between PLN and KIK.
- **Prepared and rolled out a multi-year Digital Transformation Plan** shifting IT from hardware support to digital services platform.

Sep 2015 to Dec 2020

Assistant Manager / Manager (2017)

Sembcorp Development, SG

Roles and Responsibilities

- **Carried out end-to-end project management** for self-built real estate developments across Vietnam, Indonesia and Myanmar.
- Conducted procurement, prepared tender documentation and evaluated design consultants and contractors.
- **Developed in-house masterplans and infrastructure budgetary cost estimates** in support of business development feasibility studies.
- Provided civil engineering technical oversight for utilities and infrastructure across several industrial parks.

Achievements

- **Developed the 500ha Myanmar Hlegu Industrial Park Infrastructure Masterplan** in GIS with an 8-year infrastructure cost and financial model.
- **Oversaw design and construction of 2 warehouses** (30,000m² GFA) in Hai Phong, Vietnam.
- Developed **Project Management Standard Operating Procedures** and in-house tools for real-time site progress tracking.

Sep 2012 to Sep 2015

Engineer / Senior Engineer (2014)

Surbana International Consultants, SG

Responsibilities

- Assistant Team Lead, Transport department — reporting to the Director of Overseas Infrastructure.
- Completed **Transportation Masterplans for Kigali (Rwanda), Bujumbura (Burundi), Amaravati (India), Zhongwei (China) and Changi East Industrial Zone**.
- **Designed roads and infrastructure for Changi Airport Terminal 5** pre-construction works.
- Implemented GIS-based project planning tools across active projects.

Feb 2012 to Aug 2012

River Basin Management Engineer

Halcrow / CH2M, MY

- Prepared flood mitigation schemes and **flood risk analysis for the UK and Bangkok, Thailand**; designed flood mitigation measures for a section of the River Thames.
- Standardised flood simulation workflows, **cutting preparation time from 2 hours to 10 minutes** per simulation.

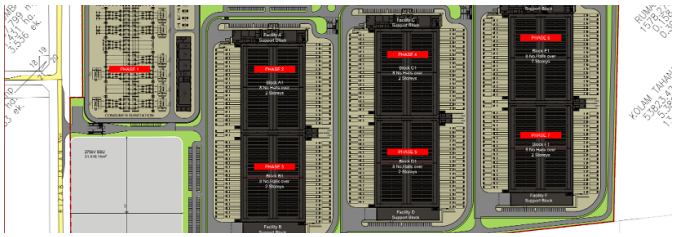
Sep 2007 to Feb 2012

Intern / Project Engineer

AKSWard Ltd, UK

- Prepared engineering design, drainage, transport and design-and-access statements, drawings and specifications for UK planning applications.
- Implemented quality management systems and ensured health and safety compliance across projects.

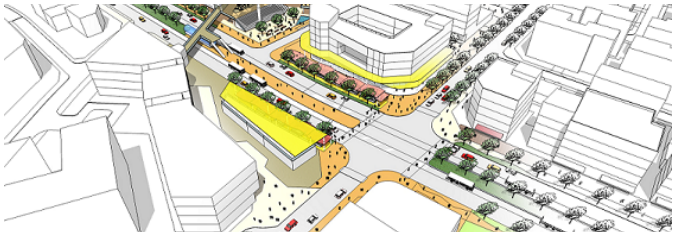
KEY PROJECTS PORTFOLIO



Concept Design & Land Acquisition — 300MW Data Centre, Johor, Malaysia
Technical Manager · Yondr Group

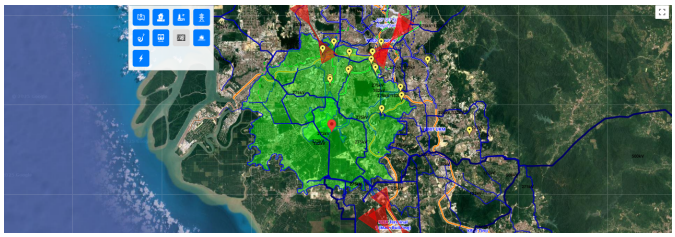
Infrastructure Design — Changi Airport Terminal 5, Singapore
Engineer · Surbana International

Infrastructure Masterplan — Myanmar Hlegu Industrial Park
Project Manager · Sembcorp Development



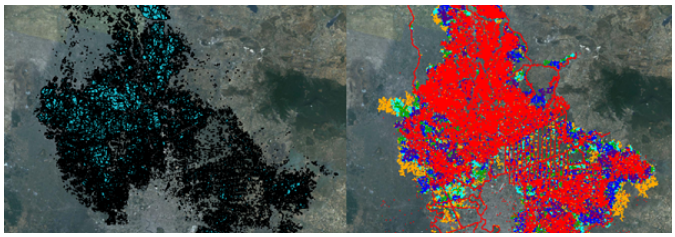
Transport Masterplanner — Kigali City Transport Masterplan, Rwanda
Co-author · Surbana International

Transport Masterplanner — Andhra Pradesh Capital City Masterplan, India
Co-author · Surbana International



Emerra.io — Geospatial site suitability platform & nine-agent AI market entry system · emerra.io
Personal Project

Real Estate Development — Phase 3 Warehouses, Hai Phong, Vietnam
Project Manager · Sembcorp Development



Flood Modelling — Dam Break, Flood Risk Analysis & River Thames Mitigation Design
Engineer · Halcrow / CH2M

SOFTWARE PROFICIENCY

Category	Tool / Platform	Proficiency
AI & LLM Systems	Claude API — multi-agent orchestration & tool use	●●●●●
	OpenAI API — Whisper, GPT	●●●●○
	LLM prompt engineering & chain design	●●●●●
	Agent SDK & agentic workflow design	●●●●○
	Python — automation, data, AI pipelines	●●●●●
	React — web application frontends	●●●●○
Geospatial & Data	QGIS — analysis, automation, plugin development	●●●●●
	ArcGIS	●●●●○
	GeoPandas / Shapely / Overpass API	●●●●○
	FastAPI — geospatial backend services	●●●●○
	Firebase / Firestore	●●●●○
PMO & Workflow	Smartsheet — API integration & automation	●●●●●
	Confluence — automated publishing	●●●●●
	Power Automate / Microsoft 365	●●●●○
	Microsoft Project	●●●●○
	LaTeX typesetting	●●●●○
Cloud & Infrastructure	Google Cloud Run — serverless backends	●●●●○
	GitHub / GitHub Pages	●●●●○
	Google Firebase — auth, hosting, Firestore	●●●●○
	SharePoint / OneDrive / Dropbox	●●●●●
	Android Studio / Flutter SDK	●●●○○
Engineering Design	AutoCAD InfraWorks / Civil3D	●●●●●
	PTV VISUM / VISSIM — transport modelling	●●●●●
	Micro Drainage / Flood Modeller — hydrology	●●●●○
	PDS Ground Modelling & Highways Design	●●●●○
	Trimble SketchUp / Revit Structure	●●●○○

EDUCATION, CERTIFICATIONS & MEMBERSHIPS

MEng in Civil Engineering
University of Portsmouth, UK
2006–2009
Distinction

BSc in Building Construction Management
Sheffield Hallam University, UK
2006
2nd Class Upper Honours

Advanced Diploma Technology (Building)
TAR College, Malaysia
2002–2006
Merit

Certifications & Memberships

Project Management Professional (PMP)	Project Management Institute
Member (MICE)	Institution of Civil Engineers, UK
Member (MCIHT)	Chartered Institute of Highways & Transportation, UK
Member	Board of Engineers Malaysia (BEM)
Member	Mensa Singapore — SIG Director (2022/2023)

References available upon request. Portfolio and live project demos at **asvinang.github.io** | Board game design · Programming · Strategy · Backpacking